Week 18
RLHF & PPO for Language Models

STUDY-RL — a self-paced Deep RL curriculum

November 8, 2026

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Learning objectives

Study companion. Read `teaching_guide.md` in this folder for step-by-step examples, an analogy with its limits, and self-checks. Then use `workbook.ipynb` to implement and vary the calculations.

Key Idea

By the end of this lecture you will be able to:

1. Derive the *Bradley–Terry* pairwise preference model and its cross-entropy loss from first principles.
2. State the *KL-regularised RL objective* for language models and explain why each term exists.
3. Derive the PPO clipped surrogate at the *token level*, including Generalised Advantage Estimation (GAE), the value head, and the adaptive reference-KL controller used in later language-model fine-tuning recipes.
4. Explain length bias, reward hacking, and reward-model overoptimisation, and pair each with a concrete mitigation.
5. Implement the full three-stage RLHF pipeline (SFT $\rightarrow$ RM $\rightarrow$ PPO) from scratch in PyTorch on a tiny transformer.
6. Read and write a TRL/PEFT recipe for the same pipeline on a Pythia-160M or SmolLM2-360M with LoRA on a single 24 GB GPU.

1 Motivation & intuition

1.1 Why classical RL is the wrong tool, naively

See the reading companion for full annotated citations.

The reader has spent seventeen weeks training agents on crisply-specified MDPs — Atari games where score changes supply reward, control suites where a sensor returns a number, grid worlds where a goal cell pays out +1. *Language is none of those things.* For an instruction-tuned model the "reward" is a fuzzy judgement ("was that a helpful response?"), the "state" is the entire prefix of text, the action space is the size of the vocabulary (often 30,000–200,000 tokens), and the trajectories last hundreds of steps. Worse, the reward is *not provided by the environment* at all: it lives inside a human reviewer's head.

The breakthrough of Ouyang et al. 2022[1] was to package this messy problem into a familiar shape. The reviewer's preferences over pairs of completions get distilled into a *neural reward model* $r_\phi(x, y)$; the language model is then a stochastic policy whose action is a token and whose state is the prefix. PPO — the same PPO that solved MuJoCo — becomes the workhorse.

Intuition. Think of RLHF as a two-step approximation. **Step one**: "what would a human say is better?" is hard, so we collect a few thousand pairwise judgements and *interpolate* them with a neural reward model. **Step two**: "what should the language model say?" is hard, so we run PPO against the interpolant, with a leash (a KL penalty) discouraging departure from the supervised baseline when the policy drifts too far into reward-hacking land.

1.2 The four-model picture

The defining ergonomic of classical RLHF is that *four* models sit in memory simultaneously (Figure 1.1). On a 24 GB consumer GPU this is uncomfortable even with LoRA — which is why the field eventually invented DPO and GRPO. We will see DPO next week.

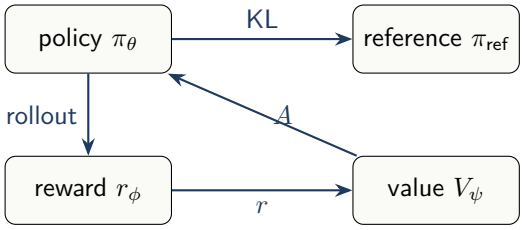


Figure 1.1: The four-model RLHF picture. The policy is trained; the reference is frozen at its SFT checkpoint; the reward model is frozen at the end of stage 2; the value head learns alongside the policy.

Common pitfall

A common confusion: "why do I need a critic if PPO has the clip?" The clip is a surrogate device that removes some incentives for large ratio changes; it is not a hard trust-region constraint and it does not estimate advantages. Without a critic, advantages are just rewards minus a moving baseline — usable, but high-variance. Token-level PPO without a

[1] InstructGPT, arXiv:2203.02155.

value head is what GRPO eventually re-introduces in week 20, with a clever group-relative baseline replacing the critic.

2 Mathematical formulation

2.1 Stage 0: the supervised fine-tuning baseline

We begin with a base language model π_{base} (pre-trained on web text). Stage 1 of RLHF is *supervised fine-tuning* (SFT): maximise the log-likelihood of N curated demonstrations $\{(x_i, y_i)\}_{i=1}^{N}$,

$$\mathcal{L}_{\text{SFT}}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{|y_i|}\log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t})\,. \tag{2.1}$$

After SFT the model is reasonably helpful but not deliberately *aligned*. We will denote the result π_{ref} — it plays a starring role in stages 2 and 3.

2.2 Stage 1: the Bradley–Terry reward model

Definition 2.2.1 (Bradley–Terry model, 1952)**.** For each prompt x and pair of candidate completions (y_w, y_l) ("winner" and "loser"), assume there exists a latent quality function $r^\star(x, y) \in \mathbb{R}$ such that the probability the annotator picks y_w over y_l is

$$\mathbb{P}(y_w \succ y_l \mid x) = \sigma(r^\star(x, y_w) - r^\star(x, y_l))\,, \quad \sigma(u) = \frac{1}{1+e^{-u}}. \tag{2.2}$$

The model is identifiable only up to an additive constant per prompt: shift $r^\star \mapsto r^\star + c(x)$ and (2.2) is unchanged.

Theorem 2.2.2 (Reward-model MLE)**.** *Given a preference dataset $\mathcal{D} = \{(x_i, y_w^{(i)}, y_l^{(i)})\}_{i=1}^{M}$ sampled i.i.d. from (2.2), the maximum-likelihood estimator of a parametric family r_ϕ minimises the cross-entropy loss*

$$\mathcal{L}_{RM}(\phi) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))]\,. \tag{2.3}$$

Proof. Each independent observation $\mathbf{1}\{y_w \succ y_l\}$ has Bernoulli likelihood $\sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$ by (2.2). The negative log-likelihood of the dataset is then exactly (2.3) up to a constant in ϕ. □ □

Remark. In practice r_ϕ is implemented as a transformer with a scalar linear head, applied to the concatenation $[x; y]$ and pooled at the last token (left-pad or attention-mask the prompt away from the loss). The two forward passes per pair, one for y_w and one for y_l, can be batched.

Common pitfall

Bradley–Terry is a *probabilistic* model with one scalar score per response. Its score ordering imposes transitive pairwise tendencies, but individual labels can be noisy. It cannot represent every cyclic population preference pattern. On separable empirical pairs, logistic likelihood may favour unbounded fitted score margins. In DPO, β is normally fixed: the fitted log-ratio margin, not the hyperparameter itself, can diverge. IPO changes that empirical objective.

2.3 Stage 2: the KL-regularised RL objective

Given a frozen reward model r_ϕ and frozen reference π_{ref}, the RLHF objective is to maximise

$$J(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\Big[\mathbb{E}_{y\sim\pi_\theta(\cdot\,|\,x)}[r_\phi(x,y)] \;-\; \beta\,\mathrm{KL}(\pi_\theta(\cdot\,|\,x)\,\|\,\pi_{\text{ref}}(\cdot\,|\,x))\Big]. \tag{2.4}$$

The KL term is a soft penalty on drift from the SFT baseline; it is not a hard upper bound on KL or a safety guarantee. A misspecified reward model can favour repetitive or otherwise poor responses. Removing the penalty can make such exploitation easier, but the behaviour depends on the reward, policy and optimisation setting.

Theorem 2.3.1 (Closed-form optimum of (2.4))**.** *For $\beta > 0$, fixed reward r, positive reference support on the considered responses, and finite $Z(x)$, the* unconstrained *maximiser of* (2.4) *over the simplex of policies $\pi(\cdot\,|\,x)$ is*

$$\pi^\star(y\,|\,x) = \frac{1}{Z(x)}\pi_{ref}(y\,|\,x)\,\exp\Big(\tfrac{1}{\beta}r(x,y)\Big), \qquad Z(x) = \sum_{y'}\pi_{ref}(y'\,|\,x)\,\exp\Big(\tfrac{1}{\beta}r(x,y')\Big). \tag{2.5}$$

Proof. Drop the prompt x to lighten notation. We maximise $\mathbb{E}_{y\sim\pi}[r(y)] - \beta\mathrm{KL}(\pi\|\pi_{\text{ref}})$ over π subject to $\sum_y \pi(y) = 1$. Form the Lagrangian

$$\mathcal{L}(\pi,\lambda) = \sum_y \pi(y)r(y) - \beta\sum_y \pi(y)\log\tfrac{\pi(y)}{\pi_{\text{ref}}(y)} - \lambda\Big(\sum_y \pi(y) - 1\Big).$$

Differentiating w.r.t. $\pi(y)$ and setting to zero,

$$r(y) - \beta\Big(\log\tfrac{\pi(y)}{\pi_{\text{ref}}(y)} + 1\Big) - \lambda = 0 \;\;\Longrightarrow\;\; \pi(y) = \pi_{\text{ref}}(y)\,\exp\Big(\tfrac{r(y)}{\beta} - 1 - \tfrac{\lambda}{\beta}\Big).$$

Normalising over y absorbs the constants $-1-\lambda/\beta$ into the partition function, giving (2.5). □ □

Remark. Equation (2.5) is the seed of *Direct Preference Optimization* (DPO). Inverting it gives $r(x,y) = \beta\log\frac{\pi^\star(y\,|\,x)}{\pi_{\text{ref}}(y\,|\,x)} + \beta\log Z(x)$, the *implicit reward* that we will use next week to bypass stages 2 and 3 entirely.

2.4 Stage 3: PPO at the token level

In practice we cannot enumerate $y \in \mathcal{A}^*$ to normalise (2.5). We instead *sample*, treating the language model as a Markov decision process whose state at step t is the prefix $s_t = (x, y_{<t})$ and whose action $a_t = y_t$ is a single token.

Definition 2.4.1 (Token-level RLHF MDP)**.** $\mathcal{S} = \mathcal{A}^*$ (all prefixes); $\mathcal{A}$ = the vocabulary; transitions are deterministic concatenation $s_{t+1} = (s_t, a_t)$; the per-step reward is

$$R_t = \begin{cases} -\beta(\log\pi_\theta(a_t\,|\,s_t) - \log\pi_{\text{ref}}(a_t\,|\,s_t)) & t < T, \\ -\beta(\log\pi_\theta(a_t\,|\,s_t) - \log\pi_{\text{ref}}(a_t\,|\,s_t)) + r_\phi(x, y_{1:T}) & t = T, \end{cases} \tag{2.6}$$

where T is the final-token index of the completion.

Remark (Why a per-token KL?). The expected sum of the per-step penalties equals $-\beta\mathrm{KL}(\pi_\theta(\cdot\,|\,x)\,\|\,\pi_{\text{ref}}(\cdot\,|\,x))$ by the chain rule of KL — so (2.6) is precisely the KL term of (2.4) unrolled across the autoregressive trajectory. Token-level credit assignment lets the value head and the advantages distinguish "it was the third word that went wrong" from "the whole thing was bad", dramatically reducing variance.

Definition 2.4.2 (Generalised Advantage Estimation, Schulman et al. 2016)**.** For a learned value estimator V_ψ and a rollout $(s_0, a_0, R_0, \ldots, s_T, a_T, R_T)$, define $\delta_t = R_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t)$. The GAE-$\lambda$ advantage is

$$\hat{A}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k \delta_{t+k}. \tag{2.7}$$

In language settings the natural discount is $\gamma = 1$ (an episode is one sequence, and discounting would change the relative value of early and late token rewards; different prefixes are different states), and $\lambda \in [0.9, 0.97]$ is standard. We bootstrap $V_\psi(s_{T+1}) = 0$.

Definition 2.4.3 (PPO clipped surrogate)**.** With $r_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ the per-token importance ratio and $\hat{A}_t$ a GAE estimate,

$$\mathcal{L}^{\text{PPO}}(\theta) = -\mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta),\, 1-\varepsilon,\, 1+\varepsilon)\,\hat{A}_t\Big)\Big]. \tag{2.8}$$

The clip ε is typically 0.2.

Theorem 2.4.4 (Why the clip is a pessimistic surrogate)**.** *For every sample with $\hat{A}_t \geq 0$, $\min(r_t\hat{A}_t, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon)\hat{A}_t)$ is a* lower bound *on $r_t\hat{A}_t$. For every sample with $\hat{A}_t < 0$ it is again a lower bound. Consequently* (2.8) *upper-bounds the negative unclipped surrogate loss. The clipped sample contributes no gradient incentive for further movement in its favourable direction; shared parameters, other samples, and other loss terms can still move its ratio farther.*

Proof. Case $\hat{A}_t \geq 0$. If $r_t > 1+\varepsilon$, the clip gives $(1+\varepsilon)\hat{A}_t \leq r_t\hat{A}_t$; the min picks the clipped value. Gradient w.r.t. θ through the clipped branch is zero in r_t, so increasing r_t further does *not* decrease the loss. Case $\hat{A}_t < 0$ is symmetric: if $r_t < 1-\varepsilon$, the clip $(1-\varepsilon)\hat{A}_t$ is again the min, gradient is zero, so this sample alone provides no gradient incentive to push r_t farther down. □ □

Intuition. The intuition behind the clip is *trust region by tape.* A proper trust region (TRPO) constrains $\text{KL}(\pi_{\text{old}} \| \pi_\theta) \leq \delta$ via a second-order line search; PPO replaces that constraint with a min / max tape on the per-sample ratio. It is much cheaper and almost as well-behaved.

2.5 The full loss

Combining the policy loss (2.8), the value loss, an entropy bonus, and the token-level KL penalty already inside the rewards,

$$\mathcal{L}_{\text{total}}(\theta, \psi) = \mathcal{L}^{\text{PPO}}(\theta) + c_v \cdot \mathbb{E}_t\Big[(V_\psi(s_t) - G_t)^2\Big] - c_e \cdot \mathbb{E}_t[\mathcal{H}(\pi_\theta(\cdot \mid s_t))]. \tag{2.9}$$

Here $G_t = \hat{A}_t + V_\psi(s_t)$ is the return target. Coefficients $c_v \approx 0.5$, $c_e \in [0, 0.01]$.

2.6 The adaptive KL controller

A reference-KL feedback controller appears in the language-model fine-tuning implementation accompanying Ziegler et al. (2019). It is distinct from PPO (2017)'s adaptive penalty on new-versus-old policy KL. One controller form is

$$\beta \leftarrow \beta \left[1 + \text{clip}\left(\frac{\widehat{\text{KL}}}{\text{KL}_{\text{target}}} - 1, -0.2, 0.2\right) \frac{n_{\text{steps}}}{H}\right]. \tag{2.10}$$

H and n_{steps} must use the same unit. The workbook helper uses one controller update per call ($n_{\text{steps}} = 1$). A target on summed sequence KL differs from a target on mean token KL, especially for varying response lengths. Feedback can oscillate and does not enforce a hard bound.

Common pitfall

With small batch sizes the observed KL is noisy and (2.10) oscillates. Two fixes: (i) accumulate KL across several rollouts before updating β; (ii) cap β in a hard range $[\beta_{\min}, \beta_{\max}]$.

2.7 A picture of token-level credit assignment



Figure 2.1: Token-level reward shaping. Each response token earns a signed reward adjustment proportional to its sampled log-ratio; the final token additionally receives the scalar reward-model score.

2.8 An empirical illustration of the BT loss

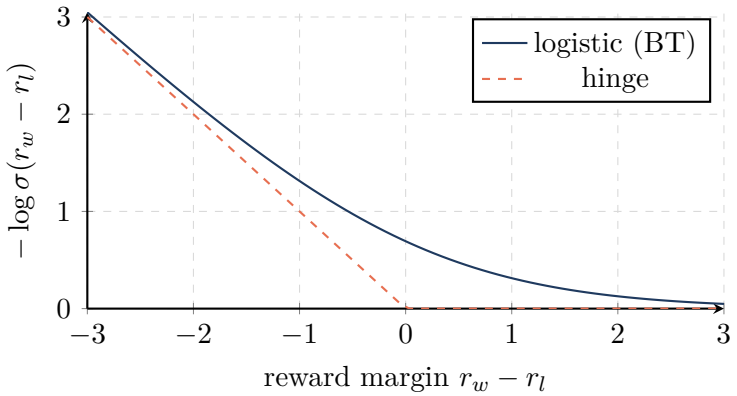


Figure 2.2: The Bradley–Terry logistic loss versus the hinge variant used by Slic-HF. At a finite real-valued margin the logistic derivative is nonzero, but it tends to zero at a large positive margin and can underflow numerically. Separable empirical pairs can favour arbitrarily large fitted margins, motivating finite-margin alternatives such as IPO.

3 Algorithms

3.1 The three-stage pipeline

Algorithm 1: Three-stage RLHF (InstructGPT-style).

Input: base model π_{base}; demos $\mathcal{D}_{\text{SFT}}$; preferences $\mathcal{D}_{\text{pref}}$; prompts $\mathcal{D}_{\text{prompt}}$
Output: aligned policy π_θ
Stage 1 (SFT): initialise $\pi_\theta \leftarrow \pi_{\text{base}}$
for *epoch* $= 1, \ldots, E_{SFT}$ **do**
  minimise $\mathcal{L}_{\text{SFT}}$ (2.1) on a batch of $\mathcal{D}_{\text{SFT}}$
end
$\pi_{\text{ref}} \leftarrow \pi_\theta$ `// freeze`
Stage 2 (RM): initialise r_ϕ from π_{ref} + scalar head
for *epoch* $= 1, \ldots, E_{RM}$ **do**
  minimise $\mathcal{L}_{\text{RM}}$ (2.3) on a batch of $\mathcal{D}_{\text{pref}}$
end
Stage 3 (PPO): initialise value head V_ψ; KL coefficient β
for *step* $= 1, \ldots, S$ **do**
  sample a batch of prompts X from $\mathcal{D}_{\text{prompt}}$
  for each prompt, sample a completion under π_θ
  compute per-token rewards (2.6) using $\pi_{\text{ref}}, r_\phi, \beta$
  compute GAE advantages (2.7) from V_ψ
  for *epoch* $= 1, \ldots, K$ **do**
    update (θ, ψ) by Adam on $\mathcal{L}_{\text{total}}$ (2.9)
  end
  update β via (2.10)
end

Complexity. Per PPO step, we run one rollout (one forward pass per token in the completion) plus K training epochs each requiring one forward+backward over the rollout. For batch B, completion length L, K=4 epochs the cost is $\mathcal{O}(B \cdot L)$ rollout-time + $\mathcal{O}(K \cdot B \cdot (P + L))$ training-time, where P is the prompt length. Peak memory also depends on activations, sequence length, frozen model storage, optimiser states and checkpointing. LoRA reduces trainable parameters and their optimiser state; it does not imply a fixed hundred-fold reduction in total model memory.

3.2 Worked numerical example

Consider a toy preference batch of two items:

$$(x_1,\, y_w^{(1)},\, y_l^{(1)}) = (\text{“cats”},\, \text{“i like cats.”},\, \text{“CATS ARE BAD!!”}),$$
$$(x_2,\, y_w^{(2)},\, y_l^{(2)}) = (\text{“tea”},\, \text{“yes, tea is nice.”},\, \text{“NO!!”}).$$

Suppose after a few RM steps the network outputs $r_\phi(x_1, y_w) = 0.8$, $r_\phi(x_1, y_l) = -0.2$, $r_\phi(x_2, y_w) = 0.5$, $r_\phi(x_2, y_l) = 0.1$. Then the BT loss (2.3) evaluates to

$$-\tfrac{1}{2}[\log \sigma(0.8 - (-0.2)) + \log \sigma(0.5 - 0.1)] = -\tfrac{1}{2}[\log \sigma(1.0) + \log \sigma(0.4)] \approx 0.41314.$$

Pairwise accuracy is 1.0 (both differences positive). After another epoch, the gaps widen to $(1.6, 0.9)$ and the loss falls to ≈ 0.26253. Compare the empirical curves in Figure 5.1.

4 Implementation walkthrough

The full project lives in project/; the highlights:

- `project/model.py` — a 2-layer 64-dim causal transformer that doubles as policy backbone and reward-model backbone. `ValueHead` and `RewardModel` bolt scalar heads onto the trunk.
- `project/sft.py` — a 30-line SFT loop.
- `project/rm.py` — Bradley–Terry training with `reward_loss` implementing (2.3).
- `project/ppo.py` — token-level rollouts, GAE, clipped surrogate, adaptive KL controller.
- `project/trl_recipe.py` — the same pipeline expressed as a TRL/PEFT script for a real LM (SmolLM2-360M).

Reward-model loss (annotated).

```
def reward_loss(r_w, r_l):
    # Bradley--Terry cross-entropy.  No reduction inside the sigmoid
    # because we want -log P(winner > loser) per pair.
    return -F.logsigmoid(r_w - r_l).mean()
```

Per-token KL-shaped rewards (annotated).

```
# kl_token = log pi_theta(a_t | s_t) - log pi_ref(a_t | s_t)
kl_token = logprobs - ref_logprobs
per_tok_rewards = -kl_coef * kl_token
# Sparse RM reward only at the final position.
per_tok_rewards[:, -1] += rm_scores
```

GAE (annotated).

```
def compute_gae(rewards, values, gamma=1.0, lam=0.95):
    B, T = rewards.shape
    adv = torch.zeros_like(rewards)
    last = torch.zeros(B, device=rewards.device)
    for t in reversed(range(T)):
        delta = rewards[:, t] + gamma * values[:, t+1] - values[:, t]
        last  = delta + gamma * lam * last
        adv[:, t] = last
    returns = adv + values[:, :T]
    return adv, returns
```

PPO step (annotated).

```
ratio = (new_logprobs - rollout.logprobs).exp()
unclipped = ratio * adv
clipped   = torch.clamp(ratio, 1-clip_eps, 1+clip_eps) * adv
policy_loss = -torch.min(unclipped, clipped).mean()
value_loss  = F.mse_loss(v_pred, returns)
loss = policy_loss + vf_coef * value_loss - ent_coef * entropy
```

Move from the toy model to the maintained notebook. `project/trl_recipe.py` is a compatibility pointer, not a pretrained PPO/RM trainer. The executable optional lab is `llm-lab/README.md` and its Jupyter workbook; it implements pretrained-model SFT and DPO with explicit dependency versions, save/reload and evaluation. The full SFT–RM–PPO loop here remains the local tiny PyTorch exercise. Do not infer hardware throughput or API compatibility from an old unpinned trainer snippet.

5 Empirical results

5.1 Statistics bridge: Twenty preference pairs may contain only two prompt units

Example 5.1.1 (Original statistical teaching fixture)**.** A reward model gets 10 of 10 preference pairs right for one prompt and 8 of 10 for another. For illustration, treat prompts as independent and compute their accuracy summaries.

The pair-weighted accuracy is $18/20 = 0.9$. Prompt accuracies are 1 and 0.8, with mean 0.9, sample SD $\sqrt{0.1^2 + 0.1^2} \approx 0.14142$, and estimated SE across the two prompt summaries 0.1. Treating all pairs as independent would instead give the plug-in Bernoulli SE $\sqrt{0.9(0.1)/20} \approx 0.06708$.

The pair-level independence premise is unjustified when outcomes share prompt-specific difficulty or labeling errors. Two prompts are too few for a robust new-prompt claim. Preference prediction and policy task performance are separate endpoints; optimizing the learned reward also changes the response distribution.

Check a changed assumption. If each prompt is duplicated ten times, what happens to the count of independent prompts? What split is needed to evaluate new-prompt behavior?

Check. The independent prompt count remains two. Split prompts before reward-model fitting and policy selection, then evaluate on a held-out prompt population matching the intended question.

Statistical foundation. Siegrist § § 4.5, 6.2, PDF pp.478, 817; Ross § 15.3, PDF p.683. Section source: https://stats.libretexts.org/@go/page/10160. The course statistics companion records the supplied editions and one-based PDF pages. This fixture is not a newly measured project result.

Figure 5.1 shows the supervised pre-training and reward-model training curves on the toy dataset; the RM reaches close to 100% pairwise accuracy in 150 steps in well under five seconds on CPU. Figure 5.2 shows the PPO reward curve: starting from ≈ 0, the average reward-model score climbs above 4 in 30 PPO updates while the adaptive KL controller keeps the empirical KL near 0.05.

Figure 5.3 is a schematic of reward-model overoptimisation in the spirit of Gao, Schulman, Hilton 2023: at small KL, proxy and gold reward agree; as KL grows, the policy "hacks" the proxy and gold reward falls.

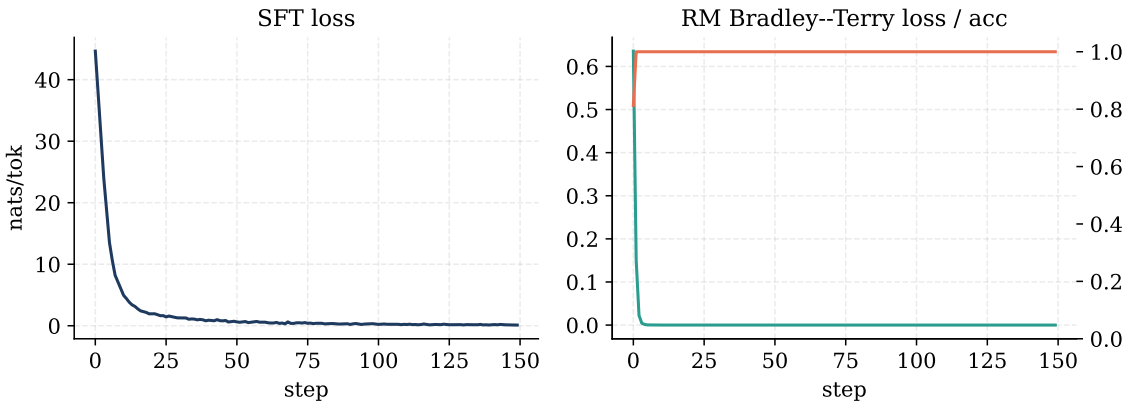


Figure 5.1: Stage 1 (SFT) and stage 2 (reward model) training curves on the toy preference dataset. Generated by `project/make_figures.py`, seed 0.

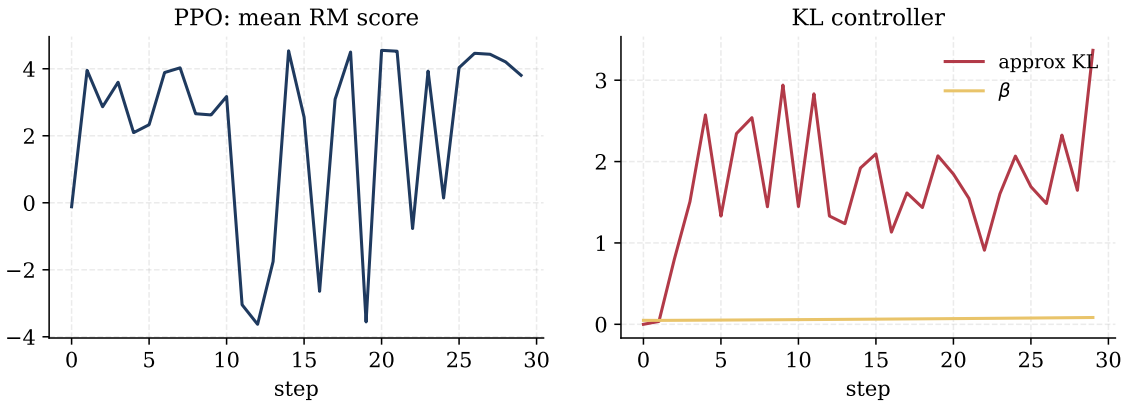


Figure 5.2: Stage 3 (PPO) reward and KL trajectories. The adaptive controller raises β when KL spikes and relaxes it when KL is under target. Generated by `project/make_figures.py`, seed 0.

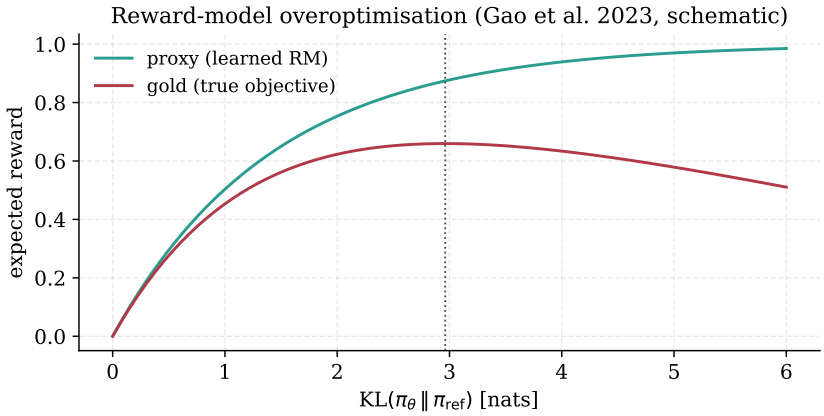


Figure 5.3: Reward-model overoptimisation (schematic). The gold reward peaks at a finite KL and then falls; the proxy keeps rising.

6 Theoretical deepening

6.1 Policy-gradient identity for autoregressive LMs

We started with the surrogate (2.8); let us anchor it in the underlying policy-gradient identity. For a stochastic policy $\pi_\theta(a \mid s)$ and an episodic return G,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \hat{A}_t\right]$$

where $\hat{A}_t$ is any baseline-corrected estimator of the advantage. Importance-sampling onto a slightly older policy $\pi_{\theta_{\text{old}}}$, the same gradient is the gradient of $\mathbb{E}_t[r_t(\theta)\hat{A}_t]$ at $\theta = \theta_{\text{old}}$. PPO's contribution is that the clipped *lower bound* on $r_t(\theta)\hat{A}_t$ has the same gradient at the on-policy point *and* removes the clipped sample's incentive to move farther in its favourable direction; it does not bound the actual update.

Lemma 6.1.1 (Local gradient equivalence at $\theta = \theta_{\text{old}}$)**.** *At $r_t = 1$, $\nabla_\theta r_t(\theta)\hat{A}_t = \hat{A}_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)$, equal to the policy-gradient estimator term. So PPO is unbiased at the on-policy point; the clip can matter once r_t leaves $[1-\varepsilon, 1+\varepsilon]$, depending on the sign of $\hat{A}_t$. Unbiasedness also requires an appropriate advantage estimator.*

Remark. At the first update on fresh on-policy data, ratios equal one irrespective of batch size. Reusing that batch after parameter updates makes it off-policy. Batch size affects sampling noise; learning rate, update count, and parameter sharing determine how far ratios move. Clipping is not a hard trust region.

6.2 Why a value head reduces variance

Without a critic the natural advantage estimator on a token-level RLHF trajectory is the *tail* of the per-token rewards: $\hat{A}_t = \sum_{k \geq t} R_k - b(s_t)$ for some scalar baseline b. The variance of this estimator is dominated by the variance of the final-token RM reward, which is *O(1)* regardless of t — so the per-token gradients all carry full-trajectory variance. GAE with a learned V_ψ cuts this variance by replacing the tail sum with $V_\psi(s_{t+1})$ wherever $\lambda < 1$:

$$\hat{A}_t = \delta_t + \gamma\lambda\delta_{t+1} + \cdots, \quad \delta_t = R_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t). \tag{6.1}$$

At $\lambda = 0$ the advantage is the one-step TD error, which is still random across sampled actions, stochastic rewards, or transitions even with a perfect V_ψ; an approximate critic can also introduce bias. At $\lambda = 1$ the advantage is the unbiased Monte-Carlo return (minus the baseline), with full variance. The sweet spot for RLHF is usually $\lambda \in [0.9, 0.97]$.

Intuition. GAE is the bias-variance dial. λ buys variance reduction at the cost of bias. The variance you remove is precisely the part of the return that should have been credited to *later* states than the current one.

6.3 Equivalence between the per-token KL and the sequence KL

We claimed in (2.6) that the sum of per-token KL penalties equals the sequence-level KL. Let us prove it.

Theorem 6.3.1 (Chain rule of KL for autoregressive policies)**.** *For any two policies π, μ on the same vocabulary and any prompt x,*

$$\mathrm{KL}(\pi(\cdot \mid x) \,\|\, \mu(\cdot \mid x)) = \mathbb{E}_{y\sim\pi}\left[\sum_{t=1}^{|y|} \log \frac{\pi(y_t \mid x, y_{<t})}{\mu(y_t \mid x, y_{<t})}\right].$$

Proof. By definition,

$$\mathrm{KL}(\pi\|\mu) = \mathbb{E}_{y\sim\pi}\left[\log \frac{\pi(y \mid x)}{\mu(y \mid x)}\right].$$

Factor each conditional via the chain rule: $\pi(y \mid x) = \prod_t \pi(y_t \mid x, y_{<t})$, and similarly for μ. The logarithm turns the product ratio into a sum that matches the right-hand side. □ □

This identity is what makes "per-token KL penalty" algorithmically honest: the expectation of the sampled log-ratio sum equals the sequence KL. One sampled sum can be negative; the KL itself cannot.

6.4 The KL controller as integral feedback in log-space

Let $e_n = \widehat{\mathrm{KL}}/\mathrm{KL}_{\text{target}} - 1$ and $\Delta_n = \mathrm{clip}(e_n, -0.2, 0.2) n_{\text{steps}}/H$. For small Δ_n, $\log \beta_{n+1} \approx \log \beta_n + \Delta_n$. Thus the log multiplier accumulates error over updates: it resembles integral feedback, not a complete proportional–integral controller. The closed-loop response also depends on policy optimisation and sampling delay. A verifier is not a replacement mathematical trust region.

6.5 Common questions, answered

Why can RLHF improve responses? Preference data can supply useful distinctions absent from demonstration-only training. Generalisation depends on label quality, coverage, optimisation and evaluation; smooth human quality near every SFT policy is not an established premise.

When might PPO and DPO differ? PPO generates new policy samples and optimises an explicit reward model; offline DPO fits fixed comparisons. Neither universally wins. Compare data budgets and independent evaluation.

Do both use a reference? Classical reference-regularised PPO and DPO both do. DPO reference log probabilities can sometimes be precomputed; this does not remove the reference from the objective.

Can SFT be skipped? Some reasoning-RL studies train from a pretrained base model. That is an empirical design choice, not proof that an arbitrary base model has enough correct samples for a sparse verifier.

Is the BT model identifiable? Only up to an additive per-prompt constant. This is fine for ranking but means you cannot compare scores across prompts.

6.6 A second worked numerical example: one PPO step end-to-end

Suppose at step 17 of PPO, the rollout for one prompt has yielded a 6-token completion with per-token log-probabilities $\log \pi_\theta = (-1.0, -1.2, -0.8, -1.5, -2.0, -1.0)$, $\log \pi_{\text{ref}} = (-1.1, -1.3, -0.7, -1.4, -1.8, -1.1)$. The per-token KL increments are $(0.1, 0.1, -0.1, -0.1, -0.2, 0.1)$, summing to -0.1 — the policy assigns this particular response lower probability than the reference. This negative sample is not a negative KL and cannot establish whether a batch-level KL target is met.

The reward model assigns the final completion $r_\phi = +0.8$. With $\beta = 0.05$, the per-token rewards are $(-0.005, -0.005, +0.005, +0.005, +0.010, +0.795)$ — nearly all the signal is at the final token, as expected.

The value head currently predicts $V = (0.0, 0.0, 0.0, 0.0, 0.0, 0.4)$ (it has learned that good rollouts end with ~ 0.8 reward but hasn't propagated credit backward yet). With $\gamma = 1, \lambda = 0.95$, the GAE TD errors are $\delta = (-0.005, -0.005, 0.005, 0.005, 0.410, 0.395)$. Starting from $\hat{A}_6 = 0.395$ and using $\hat{A}_t = \delta_t + 0.95\hat{A}_{t+1}$ gives the raw advantages $\hat{A} \approx (0.63864, 0.67752, 0.71844, 0.75099, 0.78525, 0.39500)$. Their return targets are $\hat{A} + V$; the final target is 0.795. Batch normalisation, if used, requires the full batch and cannot be inferred from this single response. For token 5, a new-versus-old log-probability increase of 0.01 gives $\rho = e^{0.01} \approx 1.01005$, so its unclipped contribution is $1.01005 \cdot 0.78525 \approx 0.79314$. With $\varepsilon = 0.2$ it is not clipped. An Adam learning rate does not bound the resulting log-probability change; inspect the actual ratios and KL after the update.

7 Safety angle

Safety angle

RLHF optimises a learned proxy. Independent evaluation should test both preference quality and failure cases not captured by that proxy.

Reward-model overoptimisation. Gao, Schulman and Hilton study proxy optimisation against a fixed larger gold model. Their fitted curves are experimental summaries, not an analytic universal law of human reward. The course schematic is not their fitted data. Compare held-out evaluations, policy drift and generated examples as optimisation continues.

Length bias. Sum versus mean losses, reward-label preferences, termination rules and length penalties can all change the weighting of responses. Neither longer sequences nor per-token averaging inevitably imply higher reward or verbosity. Report the precise denominator and measure the length–quality relationship before changing it.

Reward hacking via likelihood displacement. The policy can win at the proxy by raising the probability of a small set of high-reward strings sharing surface features with the training set — the canonical "model that ends every answer with 'I hope this helps!"' — without improving on the underlying task. *Mitigation*: KL leash; regular human spot-checks; preference-data refresh; iterative DPO (see week 19, §iterative).

Limits of controls. A KL penalty, ensemble or early stopping rule can limit a particular failure without making the reward model correct. Evaluate sycophancy, excessive refusal and factual errors separately. The scope of a theorem about reward misspecification must remain attached to its assumptions rather than being presented as a universal deployment rule.

8 Matching application

Matching application

Hypothetical matching example. Pairwise recruiter judgements about alternatives for the same context can train a preference model. A recorded click or application is not automatically a pairwise preference: exposure, availability and the alternatives shown affect its meaning. A contextual bandit or preference-optimisation experiment must define its action and logging policy. A KL penalty limits one form of deviation from a baseline; it does not establish regulatory compliance. A synthetic example can compare clicks with later retention, but no business-loss incident is asserted here.

9 Exercises

The graded version (with hidden tests) lives in `workbook.ipynb`. Short summary:

1. (⋆) Implement `reward_loss(r_w, r_l)` returning the Bradley–Terry cross-entropy.

   Intuition. Use `F.logsigmoid`, not `torch.log(torch.sigmoid(...))`; the latter is numerically unstable for large $|r_w - r_l|$.

2. (⋆) Compute the per-position KL between two logit tensors.

3. (⋆⋆) Implement the PPO clipped surrogate from scratch.

4. (⋆⋆) Implement GAE from scratch.

5. (⋆⋆) Implement the adaptive KL controller update rule.

6. (⋆⋆) Add a length-normalised reward modifier; show it reduces verbosity in the generated samples.

7. (⋆ ⋆ ⋆) Compute the unnormalised log-probability of the closed-form RLHF optimum — this is the stepping stone to week 19's DPO loss.

8. (⋆ ⋆ ⋆) *Open-ended:* build a "proxy-vs-gold" diagnostic that flags overoptimisation when the learned reward and an oracle heuristic diverge.

9. (⋆⋆) *Meaningful objective:* measure the correlation between the heuristic reward and the learned RM scores on a fresh batch of samples; report it before and after PPO to detect drift.

10 Notation, conventions, and detailed pseudocode

10.1 Notation cheat sheet

Symbol	Meaning
x	prompt (token sequence)
y	response (token sequence)
s_t	RL state $= (x, y_{<t})$
a_t	RL action $= y_t$, a single token
T	length of the response (last-token index)
$\pi_\theta(a \mid s)$	policy probability
π_{ref}	frozen reference policy (= SFT checkpoint)
$r_\phi(x, y)$	learned reward-model score (scalar, sequence-level)
$V_\psi(s)$	value head (scalar, per-token)
R_t	per-step reward (token-level, per (2.6))
δ_t	TD error $R_t + \gamma V(s_{t+1}) - V(s_t)$
$\hat{A}_t$	GAE advantage
$r_t(\theta)$	importance ratio $\pi_\theta / \pi_{\theta_{\text{old}}}$
β	KL coefficient
ε	PPO clip range (typ. 0.2)
γ	RL discount (1.0 for LLMs)
λ	GAE bias-variance dial
$\mathcal{H}$	token-level entropy

10.2 Detailed PPO rollout pseudocode

Algorithm 2: One PPO rollout batch with GAE.

Input: policy π_θ; reference π_{ref}; reward r_ϕ; value V_ψ; prompts X; max-new-tokens L; β, γ, λ

Output: advantages $\hat{A}$, returns G, rollout caches

for x *in* X *(in parallel)* **do**
- $s_0 \leftarrow x$
- **for** $t = 1, \ldots, L$ **do**
  - $a_t \sim \pi_\theta(\cdot \mid s_{t-1})$ `// ancestral sample`
  - record $\log \pi_\theta(a_t \mid s_{t-1})$, $\log \pi_{\text{ref}}(a_t \mid s_{t-1})$, $V_\psi(s_{t-1})$
  - $s_t \leftarrow (s_{t-1}, a_t)$
- **end**
- $r \leftarrow r_\phi(x, y_{1:L})$ `// single RM scalar for the rollout`
- **for** $t = 1, \ldots, L$ **do**
  - $R_t \leftarrow -\beta \left(\log \pi_\theta(a_t \mid s_{t-1}) - \log \pi_{\text{ref}}(a_t \mid s_{t-1})\right)$
- **end**
- $R_L \leftarrow R_L + r$ `// add sparse RM reward at the final step`
- `// GAE backward sweep`
- $V_{L+1} \leftarrow 0$; $\hat{A}_{L+1} \leftarrow 0$
- **for** $t = L, L-1, \ldots, 1$ **do**
  - $\delta_t \leftarrow R_t + \gamma V_\psi(s_t) - V_\psi(s_{t-1})$
  - $\hat{A}_t \leftarrow \delta_t + \gamma\lambda \hat{A}_{t+1}$
- **end**
- $G_t \leftarrow \hat{A}_t + V_\psi(s_{t-1})$ `// return target`

end

return all caches

10.3 Detailed PPO update pseudocode

Algorithm 3: Multi-epoch PPO update.

Input: rollout caches; ε, c_v, c_e, K, lr

for *epoch* $= 1, \ldots, K$ **do**
- for every position (t, b) in the batch:
  - $\log \pi_{\text{new}}(a_t \mid s_{t-1})$ from a fresh fwd pass
  - $r_t \leftarrow \exp(\log \pi_{\text{new}} - \log \pi_{\text{old}})$
  - $L_t^{\text{p}} \leftarrow -\min(r_t \hat{A}_t, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon)\hat{A}_t)$
  - $L_t^{\text{v}} \leftarrow (V_\psi(s_{t-1}) - G_t)^2$
  - $\mathcal{H}_t \leftarrow$ entropy of $\pi_\theta(\cdot \mid s_{t-1})$
- $\mathcal{L} \leftarrow \text{mean}(L^{\text{p}}) + c_v \,\text{mean}(L^{\text{v}}) - c_e \,\text{mean}(\mathcal{H})$
- $\theta \leftarrow \theta - \text{lr} \cdot \nabla_\theta \mathcal{L}$ `// Adam, grad-clipped`

end

10.4 Memory accounting on a 24 GB GPU

Memory depends on base weights, trainable parameters, optimiser state, activations, cached generation and whether value/reward models share a backbone. LoRA reduces trainable parameter and optimiser storage; it does not remove frozen weights or guarantee a fixed total-memory reduction. Measure peak memory with the actual sequence length, batch size, dtype and library

versions. The optional lab documents its measured scope; no universal 4.5 GB or 24 GB capacity claim follows from model parameter count alone.

10.5 Hyper-parameter recipe summary

Stage	Hyper-parameter	Recommended
SFT	lr	5×10^{-5} (full FT), 1×10^{-4} (LoRA)
SFT	batch	4 with grad-accum 4
SFT	epochs	1–3 over the demo set
RM	lr	1×10^{-4} LoRA
RM	batch	4–8 pairs per device
RM	target pairwise acc	> 0.7 on held-out
PPO	lr	1×10^{-6}
PPO	batch	4 prompts, grad-accum 8
PPO	rollout length	128–256 tokens
PPO	PPO epochs K	4
PPO	clip ε	0.2
PPO	γ, λ	1.0, 0.95
PPO	init β	0.05; target KL 0.1; adaptive
PPO	total steps	500–2000

11 Historical context and the road to GRPO

11.1 From REINFORCE to InstructGPT, in five papers

1. **REINFORCE** (Williams 1992). The score-function estimator $\nabla J = \mathbb{E}[\nabla \log \pi \cdot G]$. The mother of all policy-gradient methods. RLHF's gradient is REINFORCE under the hood.
2. **TRPO** (Schulman et al. 2015). Adds a hard KL trust-region constraint. Solves the policy-gradient instability problem but requires second-order methods.
3. **PPO** (Schulman et al. 2017). Replaces the trust region with a per-sample clipped ratio. First-order, works out of the box, becomes the default within a year.
4. **GPT-2 reward-model fine-tuning** (Ziegler et al. 2019). The first paper to apply PPO to a language model with a learned reward. Pioneer of the entire modern RLHF lineage.
5. **InstructGPT** (Ouyang et al. 2022). The industrial recipe: SFT $\to$ RM $\to$ PPO at scale, with adaptive KL. This describes the cited InstructGPT experiment; it does not establish an identical recipe for every later deployed product.

11.2 Why study critic-free alternatives?

A group of sampled completions supplies an alternative baseline when rewards are available at sequence level. DeepSeekMath introduces GRPO; RLOO uses a leave-one-out baseline. These change sampling, normalisation and credit assignment as well as memory use. They do not imply that PPO is invalid for terminal rewards or that token-level credit assignment is unnecessary. Week 20 derives the estimators and the limits of their unbiasedness.

11.3 Questions to carry to preference optimisation

Which supervision defines quality? Does the dataset cover the new policy? Are the reward and evaluation independent? What does the response-length normalisation change? These questions apply to PPO, DPO and critic-free methods alike. A simpler optimiser does not remove reward misspecification.

12 Further reading

See `extended_reading.pdf` for annotated summaries of Ouyang 2022 (InstructGPT), Schulman 2017 (PPO), Gao, Schulman, Hilton 2023 (scaling laws for reward overoptimisation), Bai 2022 (Constitutional AI / RLAIF), and Lambert 2024 (Tülu 3). Sourced from `research/llm_rl_sota.md` §1, §5, §6, §7.